Introduction to Machine Learning for Economic Analysis

Concepts, model fitting, out-of-sample prediction, and R practice

Gigih Fitrianto | Undergraduate Economics | Session 1

Learning outcomes: what you should be able to do

1. **Recognize** common ML tasks and major families of methods.
2. **Distinguish** prediction, explanation, and causal identification.
3. **Design** a credible modeling workflow for economic data.
4. **Run** a simple supervised prediction exercise in R.
5. **Evaluate** predictions on new data and explain bias–variance trade-offs.

Prerequisite: Basic R, data wrangling, and exploratory plots.

Three-hour workshop plan

Part I | Conceptual foundations (65 min)

- Economic questions and ML tasks
- Supervised / unsupervised learning
- ML families and modeling workflow
- Prediction, inference, and bias–variance

Part II | R demonstration (85 min)

- Read and explore an Excel dataset
- Build prior-year features
- Split by time and handle missing values
- Fit baseline, simple and multiple OLS

Short break (10 min)

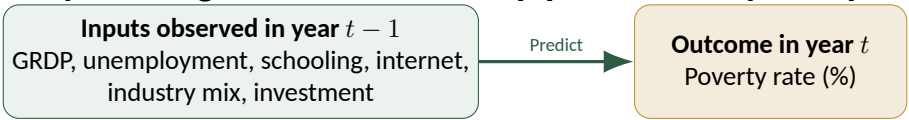
Questions and setup check

Part III | Interpretation (20 min)

- Read actual outputs
- Interpret, critique, and extend

Begin with an economic question

Can last year's regional indicators help predict next year's poverty?



The prediction question

How well does the rule predict observations it has not been trained on?

What do we mean by machine learning?

- A family of methods that **learn patterns or decision rules from data**.
- Inputs, outputs, and a performance objective depend on the task.
- Performance must be checked on data not used to estimate the rule.

$$\widehat{y} = \widehat{f}(X)$$

Learning $\widehat{f}$ is one part of a larger economic analysis workflow. Source: Mullainathan and Spiess (2017), Journal of Economic Perspectives.

Where can ML help in economic analysis?

Quantitative predictions

- Poverty or unemployment rates
- Demand or sales forecasts
- Financial or credit risk

Exploratory structure

- Regional economic clusters
- Common dimensions across indicators
- Firm or household segments

Classification

- Vulnerable / non-vulnerable households
- High / low default risk

Complex data

- Economic news and text
- Satellite or remote-sensing data

Three questions that must not be confused

Question	Economic example	Main criterion
Prediction	What will next year's poverty rate be?	Out-of-sample predictive error
Description / explanation	How are poverty and schooling associated?	Transparent patterns and assumptions
Causal identification	What is the effect of extending schooling?	Credible identification of causal effect

Important distinction

A predictive association alone does not identify the causal effect of changing a policy variable.

Econometrics and ML: a useful division of labor

Common econometric questions

- What parameter is identified?
- Which assumptions justify inference?
- How does a policy change Y?

Common predictive-ML questions

- Which observations can be predicted?
- What is error on new observations?
- Does a more complex model generalize?

Methods overlap. The research question determines the appropriate evaluation. Source: Mullainathan and Spiess (2017).

What differs: the question and the success criterion

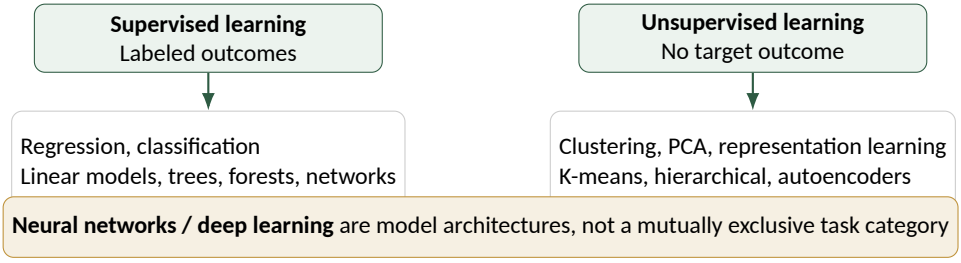
Parameter inference	Predictive modeling	Causal inference
Question: What can we learn about a coefficient?	**Question:** How accurately can we forecast an outcome?	**Question:** What would happen under an intervention?
Check: identification, assumptions, standard errors, uncertainty.	**Check:** validation/test loss and stability over time.	**Check:** causal design, counterfactual assumptions, identification.

Method and objective are different dimensions

OLS can be used for inference and as a supervised prediction algorithm. A more flexible algorithm is not inherently more accurate, and predictive associations are not automatically causal.

Source: Mullainathan and Spiess (2017).

The landscape: tasks and model families



Supervised learning: the outcome is observed

Regression

Predict a **numeric outcome**.

Example: poverty rate in percent.

Methods: OLS, LASSO, random forest.

Classification

Predict a **category**.

Example: high-risk vs low-risk household.

Methods: logistic regression, decision tree.

We will practice regression prediction today.

Unsupervised learning: explore structure without a target

Dimensional reduction

PCA summarizes correlated economic indicators using a smaller number of components.

Use: construct interpretable economic dimensions.

Clustering

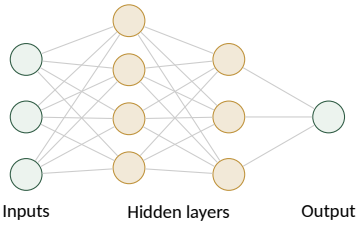
K-means and hierarchical clustering group observations with similar characteristics.

Use: characterize regional economic typologies.

Full practice will be in Session 3.

Neural networks and deep learning: where do they fit?

- A neural network combines learned transformations in layers.
- **Deep learning** uses multiple learned layers.
- It can support supervised predictions and unsupervised representations.
- Applications: complex nonlinearities, text, images, and other high-dimensional data.



Neural networks are introduced conceptually here; implementation is outside this introductory session.

Other learning settings: know the names, not the algorithms

Family	Main idea	Economic example
Semi-supervised	Combine labeled and unlabeled observations	Sparse survey labels
Reinforcement learning	Learn choices through feedback / rewards	Dynamic decisions
Self-supervised	Learn representations from the data itself	Text embeddings

Teaching scope

In this series we practice supervised prediction (Session 2) and unsupervised pattern discovery (Session 3).

Do we need Bayesian statistics and information theory?

Bayesian thinking (brief context)

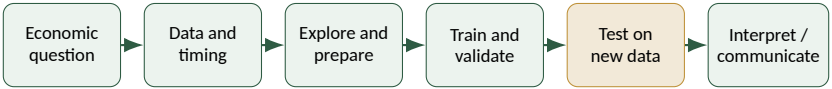
- Update beliefs using evidence
- Prior, likelihood, posterior
- Useful in probabilistic modeling and uncertainty

Information theory (brief context)

- Quantify uncertainty in information
- Entropy and information gain
- Helpful intuition for some classifiers and trees

Priority today: task definition, data timing, fitting, and out-of-sample evaluation. No derivations required.

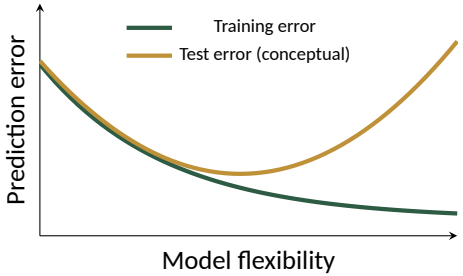
A reusable machine-learning workflow



Core rule

Evaluate how a prediction rule behaves when it is applied beyond the observations on which it was fitted.

Overfitting: why training accuracy is not enough



Schematic curves, for conceptual illustration only.

- **Underfitting:** model misses relevant structure.
- **Overfitting:** model learns unstable noise.
- **Generalization:** perform well on unseen data.

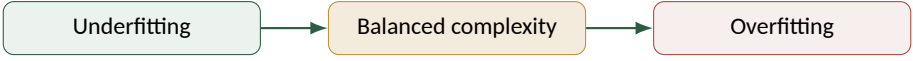
The bias–variance trade-off: an intuitive picture

Too rigid: underfitting

- Misses important economic patterns.
- **High prediction bias** is possible.
- May perform poorly in both training and testing.

Too flexible: overfitting

- Fits sample-specific fluctuations.
- **High prediction variance** is possible.
- Training fit can improve while test error worsens.



The direction and size of the trade-off depend on the true relationship, data, and model class.

Where does prediction error come from?

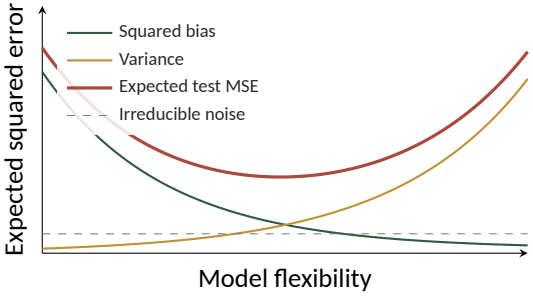
Expected out-of-sample squared prediction error at x

$$\underbrace{\mathbb{E}\left[(Y_{\text{new}} - \widehat{f}(x))^2\right]}_{\text{Expected prediction MSE}} = \underbrace{\text{Bias}\left[\widehat{f}(x)\right]^2}_{\text{Squared bias}} + \underbrace{\text{Var}\left[\widehat{f}(x)\right]}_{\text{Estimator variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}} .$$

Squared bias	Variance	Noise
Systematic deviation of the average estimated prediction from the true regression function.	How much predictions change across repeated training samples.	Unpredictable variation in the outcome, even with the correct function.

Assumptions: fixed x; $Y_{\text{new}} = f(x) + \varepsilon$, $\mathbb{E}[\varepsilon \mid x] = 0$, $\text{Var}(\varepsilon \mid x) = \sigma^2$, independent of the fitted model. Here "bias" means prediction-function bias, not omitted-variable or causal bias. Source: James et al. (2021), *An Introduction to Statistical Learning*, Ch. 2.

Why the most flexible model may not predict best



- More flexibility can reduce systematic approximation error.
- It can also make fitted predictions less stable.
- The preferred complexity minimizes **expected test loss**, not training loss.

Conceptual, schematic curves, not estimated from the workshop data. U-shape is illustrative, not universal.

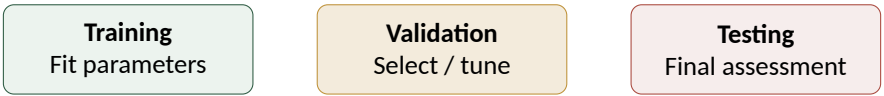
Does a more complex model outperform OLS?

Candidate	Potential strength	Possible limitation
Training mean	Transparent baseline	Ignores differences across regions
OLS regression	Transparent regression benchmark; familiar coefficients	May be unstable with many correlated predictors
Ridge / LASSO	Shrinkage may improve out-of-sample performance	Penalization introduces estimation bias; tuning required
Trees / forests	Can capture nonlinearities and interactions	Can require more data, tuning, and careful interpretation

No predetermined winner

Compare candidate models under the **same forecasting target, information set, split, and loss metric**. OLS is also a supervised learning method when used for prediction.

Training, validation, and testing: different jobs



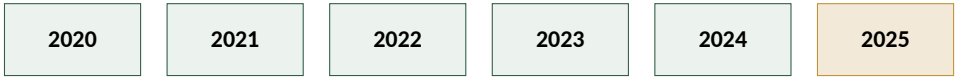
- Today's **single-model demonstration**: training period + one final holdout.
- For choosing among many models / hyperparameters, add **validation inside training**; keep the test sample untouched.

Source: Kuhn and Silge, Tidy Modeling with R; tidymodels resampling documentation.

Economic data require special care: time, groups, leakage

Risk	Example	Safer practice
Look-ahead leakage	Predict poverty in 2025 with an indicator first measured in late 2025	Use available information from 2024
Temporal dependence	Randomly mix future years into training	Hold out later years
Regional dependence	Adjacent areas share shocks	Diagnose spatial / regional patterns
Preprocessing leakage	Compute imputation using training + test	Fit imputer on training only

Our demonstration uses a time-based holdout



An ex-ante prediction

To predict poverty in 2025, the model may use measured 2024 economic indicators; 2025 poverty must remain unknown until evaluation.

How will we judge predictions?

Mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|$$

Easy to interpret: average absolute error (percentage points).

Root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}$$

More sensitive to large errors.

- **Benchmark**: predict the training-sample mean for every 2025 observation.
- **Model**: use last year's observed indicators and OLS.

Today's R workshop: three prediction rules

Rule A | Benchmark

Predict the **training mean** of poverty for every region.

No explanatory variables.

Rule B | Simple OLS

Use prior-year **schooling** to fit a line.

$$\hat{y}_{it} = \hat{\alpha} + \hat{\beta} S_{i,t-1}$$

Rule C | Multiple OLS

Use **six prior-year indicators** to predict poverty.

$$\hat{y}_{it} = \hat{\alpha} + \sum_{k=1}^{6} \hat{\beta}_k X_{k,i,t-1}$$

The teaching question

Which rule has the smallest **prediction error in 2025**, a year excluded from estimation? We compare all rules on the *same* 120 synthetic regions.

Synthetic economic data: the prediction exercise

120 fictitious regions
2020–2025
720 region-year observations

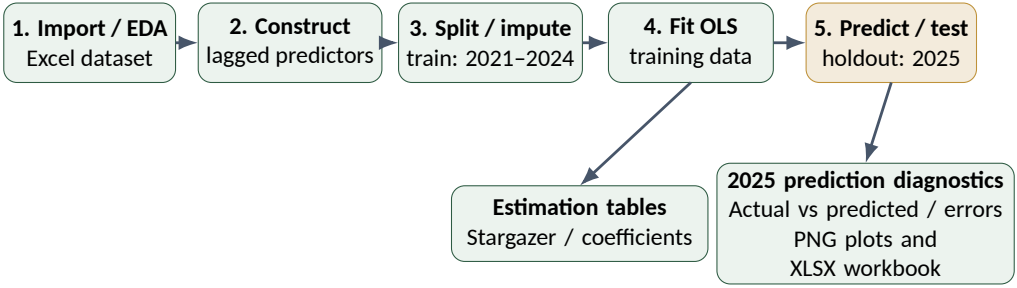
Prediction target

Poverty rate (%) in year t, using predictors observed in year $t-1$.

- GRDP per capita
- Unemployment rate
- Average schooling
- Internet access
- Manufacturing share
- Investment per capita

Source: provided ML1_Economic_Panel_Synthetic.xlsx. All observations are simulated, not measured Indonesian poverty or economic data.

From economic data to OLS prediction outputs



Interpretation of the workflow

Model coefficients are estimated on **2021–2024**; prediction errors (MAE, RMSE, and gaps) are evaluated on the **untouched 2025 holdout**. Both sets of results are exported from R.

Demo 1: import data and inspect the panel

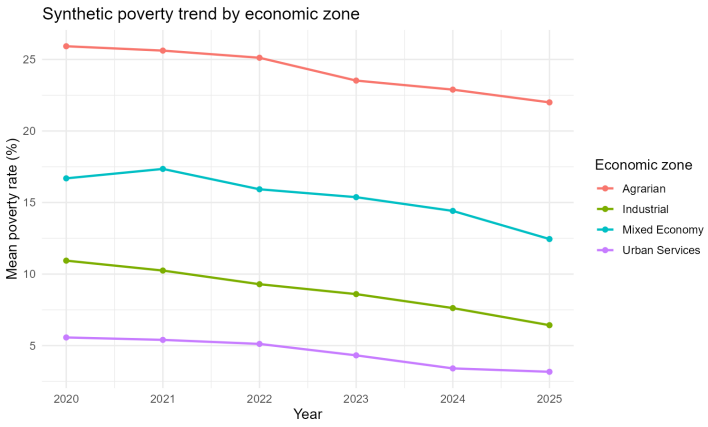
```
required_packages <- c("readxl", "dplyr", "ggplot2",
                       "stargazer", "openxlsx")
library(readxl); library(dplyr); library(ggplot2)

# In Section 0, edit input_file and out_dir for your PC.
raw <- read_excel(input_file, sheet = "Economic_Panel")
dim(raw) # 720 observations, 10 variables
count(raw, year) # 120 synthetic regions per year
n_distinct(raw$region_id)
```

Ask the class

What is an observation? Why does a six-year panel have 720 rows? Which columns identify a region and year?

Output 1: visual exploration before modeling



Actual R output: 02_poverty_trend.png.

Read the figure:

- Patterns differ by the four synthetic economic-zone groups.
- The graph describes trends in *simulated* data.
- A trend alone does not identify causal effects.

Demo 2: lag predictors within regions

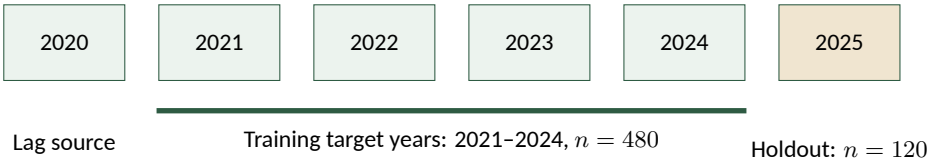
```
predictors <- c("grdp_pc_million_idr", "unemployment_pct",
  "schooling_years", "internet_access_pct",
  "manufacturing_share_pct", "investment_pc_million_idr")

panel <- raw %>%
  arrange(region_id, year) %>%
  group_by(region_id) %>%
  mutate(across(all_of(predictors), dplyr::lag,
                .names = "lag_{.col}")) %>%
  ungroup() %>% filter(year >= 2021)
```

Avoid temporal leakage

In a 2025 observation, the lagged schooling variable refers to **2024 schooling for the same region**.

Demo 3: make the training and testing split



Simple and defensible

Fit on 2021–2024; evaluate **one year ahead**, 2025. Missing-predictor medians are learned from the **training sample only** and then applied to both splits.

Demo 4: preprocessing without using test outcomes

```r
train <- panel %>% filter(year <= 2024)
test <- panel %>% filter(year == 2025)
lag_predictors <- paste0("lag_", predictors)

medians <- vapply(lag_predictors, function(v) {
  median(train[[v]], na.rm = TRUE)
}, numeric(1))
for (v in lag_predictors) {
  train[[v]][is.na(train[[v]])] <- medians[[v]]
  test[[v]][is.na(test[[v]])] <- medians[[v]]
}
```

Why this matters

Cleaning decisions made with the test outcomes can make a model appear more accurate than its genuinely out-of-sample performance.

Demo 5: estimate simple and multiple OLS

```r
# Constant prediction benchmark
baseline_value <- mean(train$poverty_rate_pct)
test$pred_baseline <- baseline_value

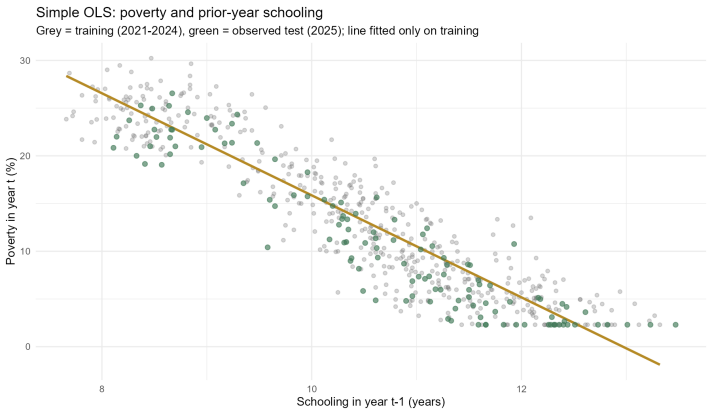
# Simple OLS: one lagged predictor
ols_simple <- lm(poverty_rate_pct ~ lag_schooling_years,
                 data = train)

# Multiple OLS: six lagged economic predictors
model_formula <- reformulate(lag_predictors,
                             response = "poverty_rate_pct")
ols_model <- lm(model_formula, data = train)
```

Interpretation

The `lm()` function estimates coefficients **on training data**. Whether those estimates predict 2025 well is a separate question.

Output 2: the fitted line is estimated on training data



Actual R output: 12_simple_ols_fitted_line.png.

Observe:

- Grey: training outcomes.
- Green: 2025 observed outcomes.
- Gold: line fitted using training *only*.

Reading the simple OLS equation

$$\widehat{Poverty}_{it} = 69.319 - 5.346\, Schooling_{i,t-1}$$

Intercept = 69.319

The estimated intercept refers to zero lagged years of schooling, a value outside the meaningful range of this demonstration. Avoid a substantive literal interpretation.

Slope = -5.346

In this synthetic training sample, an additional lagged schooling year corresponds to **5.346 percentage points lower fitted poverty** in the simple regression.

Critical distinction

This coefficient is a **predictive association**. It is not an estimated causal effect of schooling policy.

Demo 6: export regression tables using Stargazer

```r
# Display training-sample regression results in the console
stargazer::stargazer(ols_simple, ols_model,
  type = "text", digits = 3, no.space = TRUE,
  column.labels = c("Simple OLS", "Multiple OLS"))

# Save formatted tables in the output directory
stargazer::stargazer(ols_simple, ols_model,
  type = "html", digits = 3, no.space = TRUE,
  out = file.path(out_dir, "14_stargazer_ols_training.html"))
```

Two types of evidence

Stargazer reports *training-period coefficient estimates*. Holdout predictive accuracy is presented in **separate** evaluation tables.

Output 3: regression coefficients from the training sample

Variable (lagged)	Simple OLS	Multiple OLS
GRDP per capita	—	-0.080^{***}
Unemployment rate	—	0.634^{***}
Years of schooling	-5.346^{***}	-1.402^{***}
Internet access	—	-0.111^{***}
Manufacturing share	—	-0.085^{***}
Investment per capita	—	-0.242^{***}
Intercept	69.319^{***}	42.141^{***}
Observations	480	480
Training R^2	0.850	0.951

$^{***}p < 0.01$ from the conventional OLS standard errors printed by Stargazer. These figures describe simulated training data; panel dependence and causal identification are not addressed by this output.

How to interpret the two OLS specifications

1. The **schooling coefficient changes** from -5.346 in simple OLS to -1.402 in multiple OLS after including other lagged predictors.
2. The two specifications describe **different conditional associations**; their coefficients are not automatically comparable as causal effects.
3. The multiple regression has a higher **training** R^2 (0.951 vs. 0.850). This is a fit statistic on observations used to estimate coefficients.
4. The final predictive assessment requires **2025 holdout error**, which is reported separately.

Ask the class

Why does a high training R^2 not establish strong predictions on new data?

Demo 7: generate predictions for unseen observations

```r
# No re-estimation on 2025 observations!
test$pred_simple_ols <- as.numeric(
  predict(ols_simple, newdata = test))
test$pred_multiple_ols <- as.numeric(
  predict(ols_model, newdata = test))

# Signed prediction error (percentage points)
# Positive = predicted higher than actual
 test$error_multiple_ols <- test$pred_multiple_ols -
                      test$poverty_rate_pct
 test$absolute_error_multiple_ols <-
   abs(test$error_multiple_ols)
```

A genuine holdout

2025 poverty values enter here **only to score the predictions**, not to estimate model parameters.

How do we evaluate prediction error?

Mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_i |y_i - \widehat{y}_i|$$

Average size of the prediction gap in **percentage points**.

Root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \widehat{y}_i)^2}$$

Penalizes large errors more strongly.

Signed prediction error

$e_i = \widehat{y}_i - y_i$; positive means **over-prediction**, negative means **under-prediction**. Average signed error is distinct from the squared-bias term in the theoretical bias–variance decomposition.

Actual output: training and holdout performance

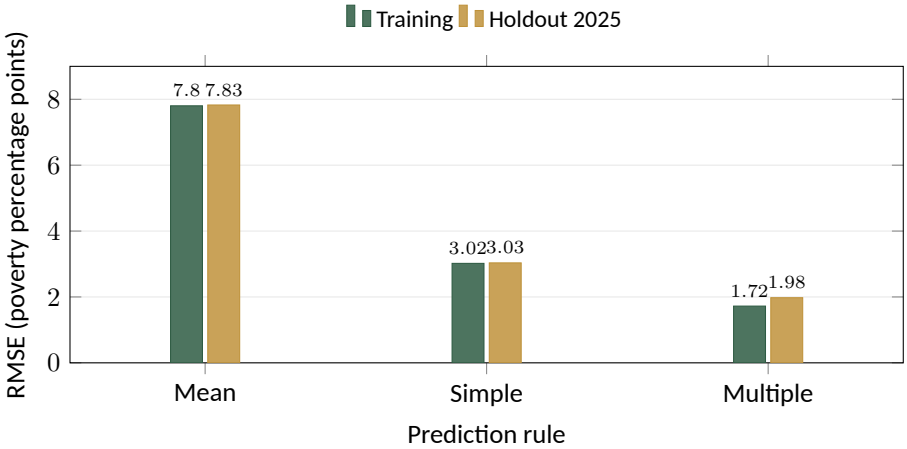
Period	Prediction rule	n	MAE	RMSE	Mean error
Training	Training mean	480	6.767	7.802	0.000
Training	Simple OLS	480	2.431	3.019	0.000
Training	Multiple OLS	480	**1.361**	**1.722**	0.000
Holdout 2025	Training mean	120	6.957	7.827	2.377
Holdout 2025	Simple OLS	120	2.439	3.034	1.283
Holdout 2025	Multiple OLS	120	**1.523**	**1.977**	0.593

What do these values mean?

Multiple OLS achieves the smallest **2025 MAE and RMSE** among the three *pre-specified* rules. In the holdout, its average absolute prediction gap is 1.523 percentage points.

Actual R output: 15_training_vs_holdout_performance.csv. Rounded to three decimals.

Comparing error: training versus 2025 testing



Computed from the supplied R output. Training and test errors are calculated on different years and sample sizes; the illustration does not establish general superiority of multiple OLS.

Why training fit and predictive performance differ

Training metrics

Multiple OLS RMSE: 1.722 pp.

The model parameters are chosen to fit this training sample. Training error is therefore optimistic as an estimate of future performance.

Holdout metrics

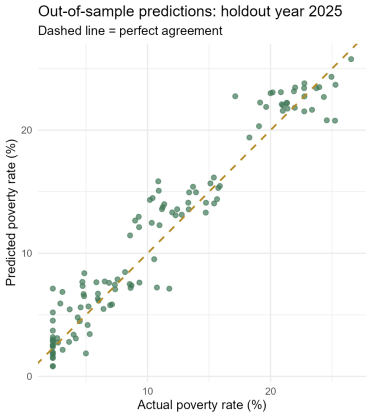
Multiple OLS RMSE: 1.977 pp.

These 2025 outcomes did not enter fitting. Prediction error rises relative to the training sample in this exercise.

A cautious comparison

This is **one synthetic holdout**. It demonstrates the logic of generalization; it does not reveal the full sampling distribution or prove that additional variables always improve out-of-sample accuracy.

Output 4: observed versus predicted poverty in 2025

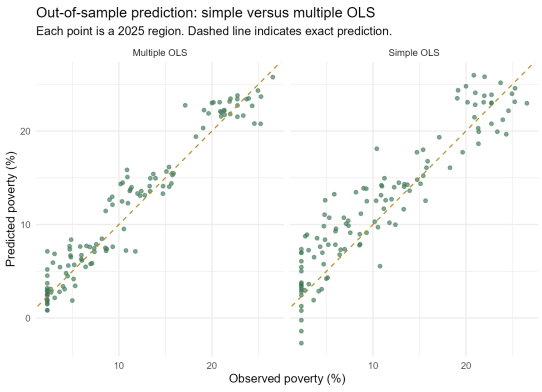


Actual R output: 08_holdout_actual_vs_predicted.png, multiple OLS, $n = 120$.

Read the diagonal:

- On the line: exact prediction.
- Above: over-prediction.
- Below: under-prediction.
- Distance signals an error.

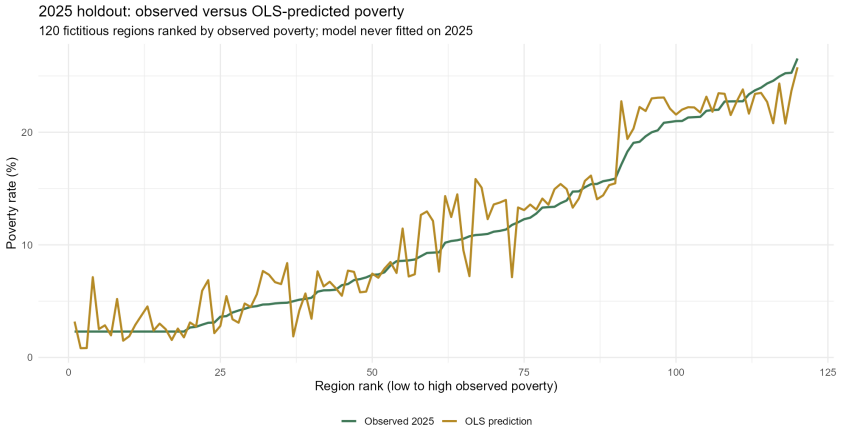
Output 5: simple versus multiple OLS on the test set



Interpretation

In this 2025 holdout, the multiple OLS points lie closer to the 45-degree line overall, consistent with its lower RMSE (1.977 versus 3.034).

Output 6: predicted and observed outcomes by region



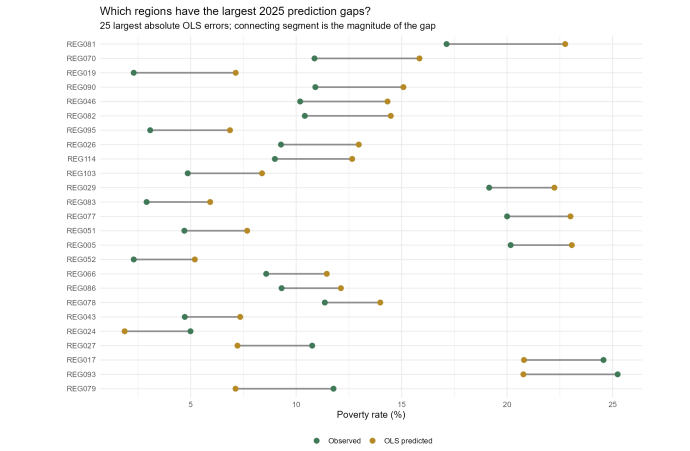
120 synthetic regions *ranked by observed 2025 poverty*, not chronological time. Gold is multiple OLS prediction; green is actual poverty.

Actual R output: 19_holdout_observed_vs_predicted_all_regions.png.

How to read the region-by-region prediction chart

1. Each horizontal position corresponds to a **fictitious region**, ordered by its observed 2025 poverty rate.
2. The **vertical gap** between the prediction and the observed line represents the signed regional prediction error.
3. The **predicted line is not forced to be monotonic**, because regions are ordered by *actual* poverty, not predicted poverty.
4. The chart shows how prediction error varies across regions. It is **not a time-series forecast chart**.

Output 7: which regions have the largest gaps?

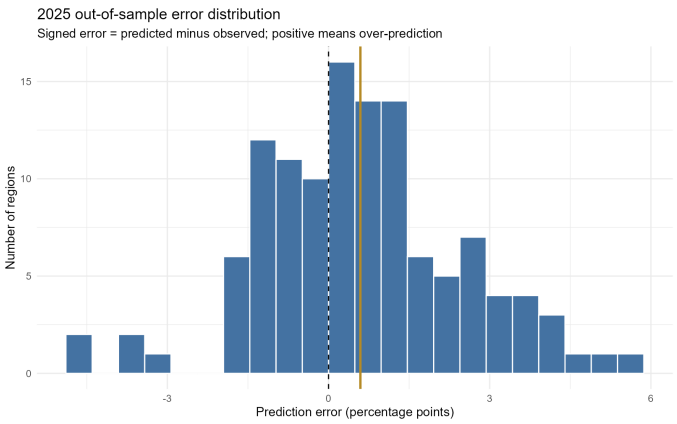


Dumbbell chart:

- Green dots: observed values.
- Gold dots: predictions.
- Longer connecting lines: larger absolute errors.
- Plot shows the **25 largest gaps only**.

Actual R output: 20_holdout_prediction_gaps_top25.png. Full $n = 120$ is retained in the Excel predictions sheet.

Output 8: signed prediction errors in 2025



Definition: predicted − observed.

- Positive: model predicts too high.
- Negative: model predicts too low.
- Mean error: **+0.593 pp**.

Actual R output: 21_holdout_prediction_error_distribution.png. Average signed error is not the bias term of the theoretical repeated-sampling decomposition.

Output 9: errors differ across economic-zone groups

Synthetic zone	n	OLS MAE	OLS RMSE	Mean error
Agrarian	30	1.588	2.054	0.472
Industrial	30	1.666	2.133	0.587
Mixed Economy	30	**1.908**	**2.328**	1.538
Urban Services	30	0.930	1.204	-0.227

Interpretation

Multiple OLS has the highest average error in the fictitious **Mixed Economy** group and the lowest in **Urban Services**. Aggregate test RMSE alone can conceal subgroup variation.

Actual R output: 23_zone_diagnostics_extended.csv. All metrics are based on the 2025 holdout.

Demo 8: save all predictions to a single workbook

```
# After producing model metrics and regional predictions:
wb <- openxlsx::createWorkbook()
# The script writes these sheets with writeDataTable():
# Overview, Model_Performance, Predictions_2025,
# Training_Fitted, OLS_Coefficients, Zone_Diagnostics,
# Training_Medians, Plots.

# It also inserts PNG charts into the Plots worksheet.
openxlsx::saveWorkbook(
  wb, file.path(out_dir,
      "24_ML1_OLS_Prediction_Results.xlsx"),
  overwrite = TRUE)
```

Teaching value

Students can inspect **120 individual test forecasts**, compare methods, and trace the results back to their predictors without having to re-run every plot.

What is actually in the output workbook?

Worksheet	Purpose
Overview	Sample, date split, prediction task, interpretation caveats
Model_Performance	Train and test MAE, RMSE, mean error and R^2
Predictions_2025	Observed, forecast and regional errors for all 120 regions
Training_Fitted	Fitted training-period outcomes and errors
OLS_Coefficients	Estimate, standard error, t-statistic and p-value
Zone_Diagnostics	Error indicators for four fictitious economic groups
Training_Medians	Training medians used for imputation
Plots	Embedded observed-predicted and regional-gap figures

Verified from supplied file: 24_ML1_OLS_Prediction_Results.xlsx.

Where to find the research outputs

Tables and workbook

- **13–14:** Stargazer, TXT and HTML
- **15:** Training and test performance
- **16–17:** Regional predictions and fitted values
- **23:** Errors by synthetic economic zone
- **24:** Eight-sheet results workbook

Actual figures (PNG)

- **12:** Simple OLS fitted line
- **19:** Observed vs. predicted, all regions
- **20:** Top 25 prediction gaps
- **21:** Signed error distribution
- **22:** Simple vs. multiple OLS

Output folder

Each numerical prefix corresponds to a file in `ML1_outputs v2`. The full filenames are recorded in the instructor's R code and provided ZIP archive.

What can these outputs establish, and what remains open?

Supported by this exercise

- A simple workflow produces 2025 out-of-sample predictions.
- Multiple OLS has the lowest holdout error among the three shown rules.
- Signed error and subgroup analysis reveal systematic differences within the *synthetic* dataset.

Not established

- A causal effect of schooling or other predictors.
- General superiority of multiple OLS, Ridge, or LASSO.
- Reliable prediction for actual Indonesian regions or future years.

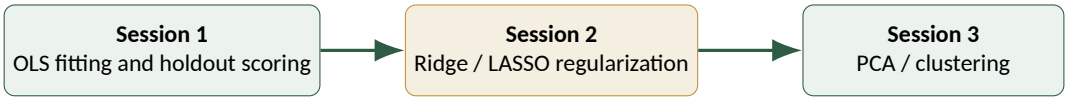
Model selection caution

The comparison is instructional. If repeated model choices are made using the 2025 holdout, a fresh final test period is needed to assess the selected model.

Classroom exercises: interpret the fitted output

1. Explain why **Simple OLS's** schooling coefficient differs from the schooling coefficient in **Multiple OLS**.
2. For the 2025 holdout, compare **MAE** and **RMSE**. Which method has the smallest value? By how much?
3. Identify whether **positive signed prediction error** means over- or under-prediction.
4. Open the `Predictions_2025` sheet. Find two regions with large absolute errors. Explain how the gap figure displays them.
5. Why should we be cautious about comparing future performance using a **single test year**?

Connecting this workshop to Ridge and LASSO



Next statistical question

Can penalizing the magnitude of coefficients reduce variance and improve unseen-data predictions, even if training fit becomes worse? This is where **Ridge and LASSO** enter.

Key takeaways from the R results

1. **Model fitting and prediction are different stages.** OLS coefficients are estimated on training data; performance is checked on 2025 holdout outcomes.
2. **Benchmarking matters.** A simple training mean provides a point of comparison for test MAE and RMSE.
3. **Actual-versus-predicted plots reveal patterns** that aggregate metrics cannot show.
4. **Subgroup errors matter.** Predictive quality can differ substantially across economic groups.
5. **No causal conclusion follows** from a predictive regression of synthetic poverty outcomes on lagged indicators.

References and self-study

1. Mullainathan, S., & Spiess, J. (2017). Machine Learning: An Applied Econometric Approach. *Journal of Economic Perspectives*, 31(2), 87–106. https://doi.org/10.1257/jep.31.2.87

2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: With Applications in R* (2nd ed.). Springer. https://www.statlearning.com/

3. Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). https://r4ds.hadley.nz/

4. Kuhn, M., & Silge, J. *Tidy Modeling with R*. https://www.tmwr.org/

Workshop empirical illustrations are based exclusively on the synthetic R outputs supplied for this session (October 2026).